Engineering Design, Design Constraints

DAEN 459 / Week 10 / Spring 2025

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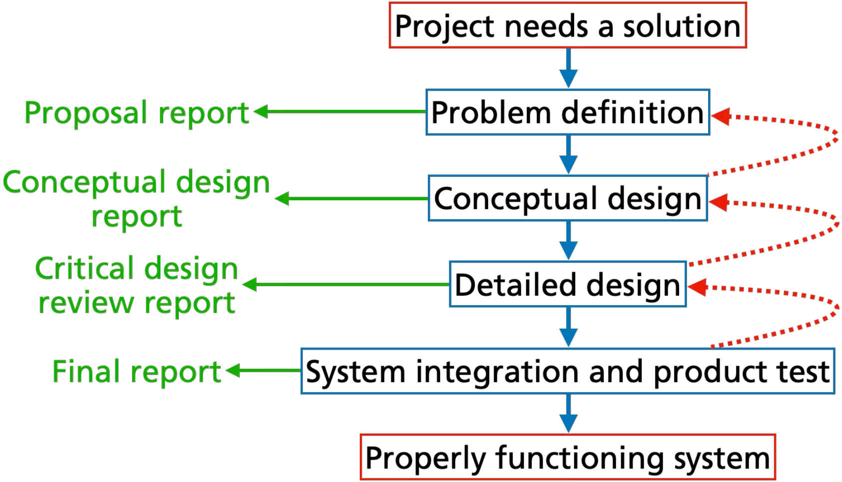
Engineering design is a systematic, iterative problem-solving process used to create products, systems, or solutions that meet human needs.

Design Process



Fundamentals of Engineering Design

While approaches vary, the core fundamentals typically include:

1. Problem Definition
2. Research & Information Gathering
3. Idea Generation (Conceptual Design)
4. Concept Selection
5. Detailed Design & Modeling
6. Prototype Development
7. Testing & Evaluation
8. Iteration & Improvement
9. Final Design & Implementation
10. Communication

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Senior Capstone Planning *(Fall)*

1. **Problem Definition**
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4. **Concept Selection**

Senior Capstone (Spring)

5. Detailed Design & Modeling
6. Prototype Development
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10. Communication

Senior Capstone Planning

1. **Problem Definition**

   Understand the need or challenge

   Identify stakeholder requirements

   Establish design goals and constraints (cost/safety/regulations/sustainability)

2. **Research & Information Gathering**

   Review existing solutions and technologies

   Benchmark standards and materials

   Identify limitations and opportunities

3. **Idea Generation (Conceptual Design)**

   Brainstorm multiple concepts

   Use creativity tools (sketching, mind mapping, morphological charts)

   Avoid premature judgment – explore alternatives

4. **Concept Selection**

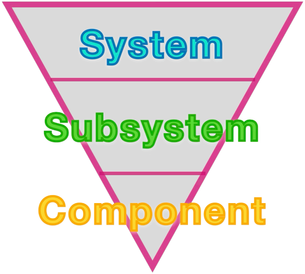
   Evaluate concepts based on criteria (feasibility/cost/performance)

   Use decision matrices or trade-off analysis

   Select the most promising design

Problem Definition

- Assessment of Needs
- Define top-level functional requirements
- Define objectives and metrics
- Specify performance requirements
- Identify constraints



Ref. Ralph M. Ford and Chris S. Coulston, **Design for Electrical and Computer Engineers: Theory, Concepts and Practice**, Learning Solutions, 2005.

Assessment of Needs

The aim is not to solve the problem
but to understand what the problem is.

- What does this client want?
- What is the problem that the design is to solve?

Functional Requirements

Specifies a behaviour that a system or subsystem must perform.

- Expressed as ”doing” statements.
- Typically involve output based on input.

Example

Design and construct a robot which can compete with a similar robot in pushing egg shaped "balls" along a playfield and place them in "nests" assigned for them, before the opponent.

Example

- *Detect start signal.*
- *Detect the egg.*
- *Detect the nest.*
- *Align the robot, the egg and the nest.*
- *Push the egg towards the nest by controlling it.*
- *Place the egg into the nest.*

Define Objectives

- Objectives, are the desired attributes of the design, what the design will "be" and what qualities it will have.
- They are often adjectives/adverbs (e.g., fast, low cost).

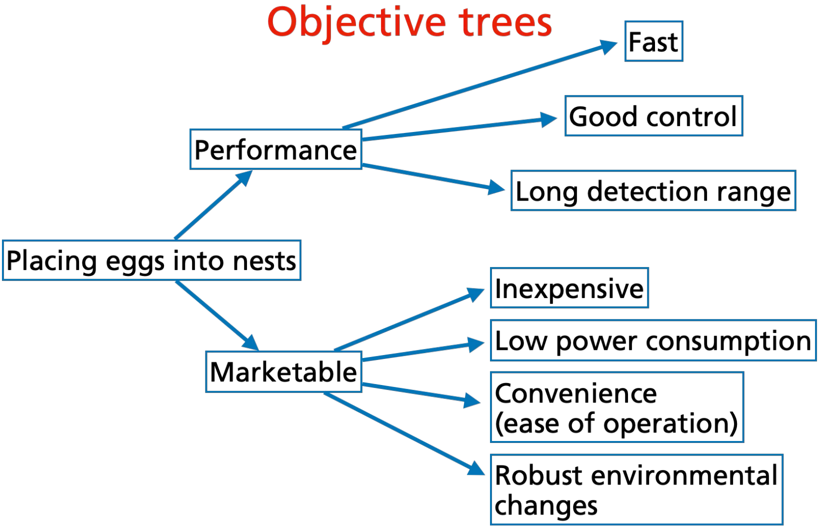
Objective Examples

- Performance related
  - Speed
  - Accuracy
  - Resolution
- Cost
- Ease of use
- Reliability, durability
- Power
  - Voltage levels
  - Battery life

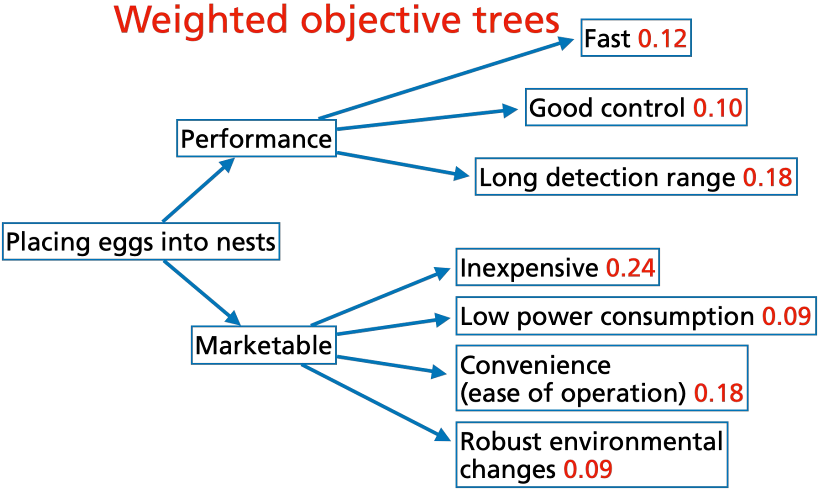
Objective Trees

- Make a list of objectives.
- Group the relevant objectives.
- Form a hierarchical tree structure.

Example


Objective trees
Fast
Good control
Performance
Long detection range
Placing eggs into nests
Inexpensive
Low power consumption
Marketable
Convenience
(ease of operation)
Robust environmental
changes

Example


Weighted objective trees
Placing eggs into nests
Performance
Fast 0.12
Good control 0.10
Long detection range 0.18
Marketable
Inexpensive 0.24
Low power consumption 0.09
Convenience (ease of operation) 0.18
Robust environmental changes 0.09

Why do we need objectives?

- Objectives allow exploration of the design space to choose among alternative design configurations.
- Three design alternatives

  Design 1: ...

  Design 2: ...

  Design 3: ...
- Which one is the best choice according to my objectives?

Evaluation of Design Alternatives

- Define objective metrics.
- Assign scoring criteria.
- Apply ratings to each design alternative.
- Compare results to support selection.

Example *Metrics indicate how well each objective is met*
*rated on a 10-point scale (**10** = Excellent, **0** = Failure).*
*No metric may be rated as **0**, or below **3**,*
the alternative with the highest average score is selected.

Specify performance requirements

- A requirement specifies a capability or a condition to be satisfied.

  Expressible as numbers and measures.

  Capability: *Works in the dark and under sunlight.*

  Condition: *Operation time <30 min.*

- Translates needs into terminology that helps us to measure how well we met them.

  It turns the problem statement into a technical, quantified form.

Requirement Types

- Functional

  Defines what the system must do to meet its intended purpose.
- Performance

  Quantitatively specifies how well the system or component must perform.
- Physical

  Specifies the physical characteristics of the system or component (e.g., weight, size, materials).

A Good Requirement Is:

- Abstract

  States what the system must do, not how it will be implemented.
- Unambiguous

  Clearly interpreted in only one way.
- Traceable

  Directly linked to the user's needs and project objectives.
- Verifiable and measurable

  Can be objectively tested to confirm it is met.

  Requires planned verification methods (test plan!).
- Achievable (realistic, feasible)

  Supported by engineering knowledge, data, and available resources.

Relation Between Requirements and Test Plans

- What is a requirement?

  What the system must do, not how it will be implemented.

- Example requirement:

  The robot shall detect a 5 kHz sine wave emitted by a mobile phone.

- What does the test plan specify?

  How far will the phone be located?

  What are the environmental conditions?

- Example testable requirement:

  The robot shall detect a 5 kHz sine wave emitted by a mobile phone placed 1 m away, under a signal-to-noise ratio of 20 dB.

Examples of good requirements

1. *The robot must have an average forward speed of 0.5 feet/sec, a top speed of at least one foot/sec, and the ability to accelerate from standstill to the average speed in under one second.*
2. *The robot should place the first egg in the nest within at most 20 min.*

Examples of poor requirements

1. The computer shall process & display the radar information instantly.
2. The ship shall carry enough short range missiles.
3. The aircraft shall use stainless steel rivets.
4. The power supply output shall be 28 volts.
5. The power supply unit shall provide 12 V DC with a load regulation of 1% while the line voltage variation is 220 +/- 20 V AC under all load current regimes and vibration and shock profiles within the temperature range.

Why??

Examples of poor requirements

1. The computer shall process & display the radar information instantly.
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Why?? In general, these violate fundamental principles of good system requirements: **clear, testable, feasible, unambiguous, and solution-independent**.

Examples of poor requirements

The computer shall process & display the radar information instantly.

- A good requirement would specify maximum latency or update frequency.

The ship shall carry enough short range missiles.

- "Enough" is ambiguous – how many is enough? Replace with a quantified, testable number tied to a mission requirement.

The aircraft shall use stainless steel rivets.

- This is a design solution, not a functional requirement.

The power supply output shall be 28 volts.

- Better to specify interface standards or compliance ranges.

The power supply unit shall provide 12 V DC with a load regulation of 1% while the line voltage variation is 220 +/- 20 V AC under all load current regimes and vibration and shock profiles within the temperature range.

- Looks detailed but still problematic. Should be separated into multiple single, measurable requirements.

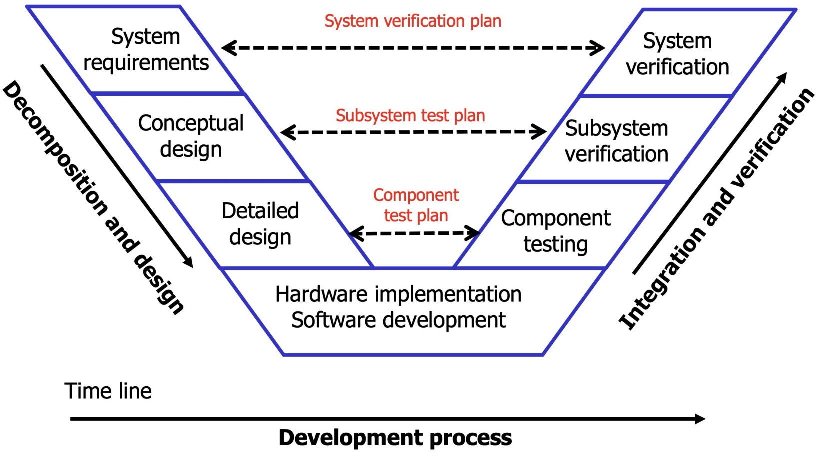
Identify Constraints

- Restrictions or limitations on a behavior, value, or aspect of performance.
- Stated as clearly defined limits.
- Often derived from guidelines, standards, or external conditions.

Example constraints for the egg-placing project:

1. *Maximum size of the robot, pushing plate, and nest.*
2. *Defined visual markers for robot and nest detection.*
3. *Start signal: 5 kHz sine wave.*

V Diagram



System requirements
Conceptual design
Detailed design
Hardware implementation
Software development
Component testing
Subsystem verification
System verification
System verification plan
Subsystem test plan
Component test plan
Decomposition and design
Integration and verification
Time line
Development process